Institute of Information technology
Jahangirnagar University
3rd year 1st Semester B.Sc.(Hons.) Final Examination 2023

Course Code: ICT-3101 | Course Title: Operating Systems
Marks: 60 | Time : 03 Hours

Part: A

Note: Answer **ALL** of them from Part A. Course Outcome (CO) and Mark of each question are mentioned at the left and right margin respectively.

(CLO1) 1. (a) Explain Bootstrap program using figure. Identify several advantages and disadvantages of open-source operating systems. 3

(CLO2) (b) How does an interrupt differ from a trap? Discuss the interrupt-driven I/O cycle with figure. 3

(CLO3) (c) Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 6

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total seek distance(in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?

i. FCFS ii. SSTF iii. SCAN iv. LOOK v. C-SCAN vi. C-LOOK

Part: B

Note: Answer any Four (4) of the following questions from Part B. Mark of each question are mentioned at the right margin respectively.

2. (a) Explain process scheduling and how it maintains different scheduling queues of processes? Represent process scheduling using queueing diagram. 3

(b) How does process cooperation enhance convenience for users? Differentiate between **message-passing** and **shared-memory** models of IPC. 3

(c) Consider the following set of processes, with the length of the CPU burst time given in milliseconds: 6

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

i. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).

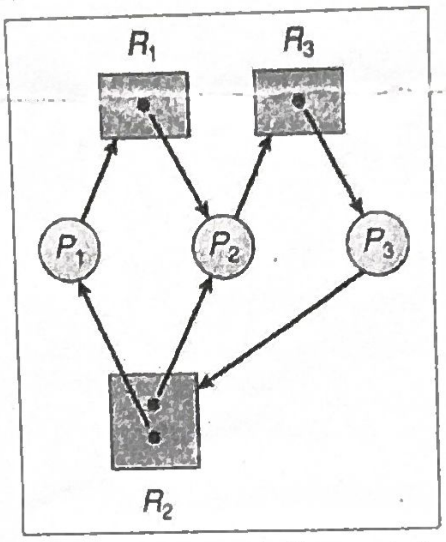
ii. What is the turnaround time of each process for each of the scheduling algorithms?

iii. What is the waiting time of each process for each of these scheduling algorithms?

iv. Which of the algorithms results in the minimum average waiting time (over all processes)?

(a) What are the benefits of using threads in a multithreaded environment? Discuss the disadvantages of user threads compared to kernel threads. 3

(b) How do you determine deadlock in a resource allocation graph? Does the following resourse allocation graph contain a deadlock? Justify your answer. 3



2
1
8
4
5

(c) Assume that there are 5 processes, P_0 through P_4, and 4 types of resources. At T_0 we have the following system state: 6

Max Instances of Resource Type A = 3 (2 allocated + 1 Available)
Max Instances of Resource Type B = 17 (12 allocated + 5 Available)
Max Instances of Resource Type C = 16 (14 allocated + 2 Available)
Max Instances of Resource Type D = 12 (12 allocated + 0 Available)

Given Matrices												
	Allocation Matrix (N0 of the allocated resources By a process)				**Max Matrix Max resources that may be used by a process**				**Available Matrix Not Allocated Resources**			
	A	**B**	**C**	**D**	**A**	**B**	**C**	**D**	**A**	**B**	**C**	**D**
P_0	0	1	1	0	0	2	1	0	1	5	2	0
P_1	1	2	3	1	1	6	5	2				
P_2	1	3	6	5	2	3	6	6				
P_3	0	6	3	2	0	6	5	2				
P_4	0	0	1	4	0	6	5	6				
Total	**2**	**12**	**14**	**12**								

i. Use the safety algorithm to test if the system is in a safe state or not?

ii. If the system is in a safe state, can the following requests be granted, why or why not? Please also run the safety algorithm on each request as necessary.

a. P1 requests (2,1,1,0)
b. P1 requests (0,2,1,0)

(a) Consider following page reference string: 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1. Compare the performance for the following page replacement algorithms assuming three frames. 4

i. Optimal Algorithm
ii. Least Recently Used Algorithm

(b) Define page fault? Discuss the steps in handling a page fault with figure? What Happens if there is no Free Frame? 4

(c) Mention the causes of thrashing? How does the system detect thrashing? Considering the following segment table and page table, translate the following logical address to physical address: (2, 6), where page size=2. 4

Segment Table

Segment	Limit	Page Table Base
S0	10	1000
S1	14	1200
S2	12	1400

Page Table

P0	1000	F3
P1	1001	F2
P2	1002	F1
P3	1003	F0

5. (a) Show with an example how the size of main memory increases the CPU utilization ? 3

(b) Compare internal fragmentation with external fragmentation. To minimize the probability of external fragmentation what we can use? Discuss the process . 3

(c) How would first-fit and best-fit algorithms place the following set of processes in order? 6

Process	Size	Turnaround	Block	Size
P1	100k	1-3-2-1	B1	50k
P2	10k		B2	200k
P3	35k	2-2	B3	70k
P4	15k	1	B4	115k
P5	23k	2-2	B5	15k
P6	6k			
P7	25k			
P8	55k	1-2-1-1		
P9	88k	1-3-1-1		
P10	100k	1-3-1-1		

6. (a) List three examples of deadlocks that are not related to a computer system environment. 3

(b) Describe briefly the process of recovery from deadlock by process termination and resource preemption. 3

(c) Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. 6

a. How many bits are there in the logical address?
b. How many bits are there in the physical address?

7. (a) How address binding of instructions and data to memory addresses can happen? 2

(b) OS is responsible for the protection of user's memory. This is accomplished by hardware. Draw the hardware and describe its operation principle. 6

(c) Describe the general dynamic storage allocation problem which concerns how to satisfy a request of size n from a list of free holes. 4

INSTITUTE OF INFORMATION TECHNOLOGY

JAHANGIRNAGAR UNIVERSITY

3RD YEAR 1ST SEMESTER BSC IN INFORMATION AND COMMUNICATION TECHNOLOGY

Final Examination 2023
Course Code: IT 3103
Course Title: Computer Network

Time Allowed: 3 Hours
Full Marks: 60

Do not write anything in the question script.
There are seven questions.
Answer any five of them including Question 1 (Compulsory).
Figures in the right margin indicate marks.

1. (a) [CO1, CO2] In our classes, we discussed various link-state and distance-vector routing protocols. You are now tasked with selecting the most suitable routing protocol for the following network scenarios, justifying your choice with a detailed explanation: [3]

   Case 1: Large Enterprise Network

   This network encompasses a complex topology with numerous interconnected devices and high bandwidth requirements to support critical applications like voice, video conferencing, and high-speed data transfer.

   Case 2: Small Branch Office

   This network consists of a small branch office with *limited bandwidth and a relatively simple network topology.*

   (b) *[CO1, CO2] Describe the roles of RTP, RTCP, and SIP protocols in enabling multimedia communication within computers or end nodes.* [3]

   (c) *[CO3]* A network administrator is tasked with designing a subnetting scheme for a network with three LANs connected by two WAN links. The ISP has allocated the IP address block 192.168.1.0/24 to the network. [6]

   HQ: Requires 50 hosts
   RO1: Requires 30 hosts
   RO2: Requires 10 hosts

   (i) Determine the minimum subnet mask (in CIDR notation) required for each LAN to accommodate the necessary number of hosts.

   (ii) Assign a unique subnet address to each LAN, ensuring efficient utilization of the allocated IP address space.

   (iii) Calculate the number of usable host addresses in each subnet.

2. (a) Explain how cookies can improve website performance by speeding up page loads. How can an excess of cookies degrade browsing speed and potentially impact your computer's overall performance? [6]

   (b) Explain how web caching improves web performance. Discuss the different types of web caches (e.g., browser cache, proxy cache) and their respective roles. Analyze the impact of caching on server load, network traffic, and user experience [3]

   (c) Let the value of the RTT is to be measured. The value of previous RTT is $350\mu s$. Let the value of length is 90%, calculate the new RTT and the transmission time. Assume that it takes a segment at this moment to be acknowledgement is $90\ \mu s$ [3]

3. (a) Explain how P2P file sharing protocols (e.g., BitTorrent) and streaming stored video protocols (e.g., protocols used by YouTube) are implemented. Discuss the key technologies involved, such as peer-to-peer networking, content delivery networks (CDNs), and adaptive bitrate streaming, and provide examples of their use in real-world applications. [6]

(b) The main challenge of VoIP is to recover from packet loss given small tolerable delay between original transmission and placback. How does the P2P architecture in Skype scale to accommodate a massive and growing user base?Consider the role of supernodes, distributed hash tables (DHTs), and other techniques in maintaining efficient peer discovery and connection establishment [6]

4. (a) Suppose you are sending an email from your gmail account to your friend, who reads his/her e-mail from his/her mail server using IMAP. Briefly describe how your email travels from your host to your friends host. Explain it using illustrations. Also, what are the application layer protocols involved? [6]

(b) In what way is instant messaging a hybrid of client-server and P2P architecture? [3]

(c) Describe the message format and the message transfer and the underlying protocol involved in the working of the email [3]

5. Imagine a medium-sized enterprise with three interconnected departments: Sales, Marketing, and IT. Each department has its own local area network (LAN) connected via routers. The Sales and Marketing departments are located in the same building, while the IT department is in a separate building. The network administrator wants to ensure efficient and reliable communication between all departments.

(a) Explain the detailed process of how an OSPF router within this network determines the best path to a destination network. Discuss the factors that contribute to the OSPF metric and how they influence route selection. Analyze potential challenges and limitations of using OSPF in this specific network scenario. [5]

(b) Describe the mechanism by which an EIGRP router within this network selects the most optimal path to a destination network. Explain the significance of the various factors considered in the EIGRP metric calculation (bandwidth, delay, load, reliability, MTU). Evaluate the advantages and disadvantages of using EIGRP in this enterprise network environment. [5]

(c) Consider a scenario where the Sales department needs to communicate with the IT department. The router in the Sales department will use its routing table to determine the best path to the IT department's network. If both EIGRP and OSPF are configured, the router will compare the metrics of the paths learned from both protocols and select the path with the lowest metric. [2]

6. (a) Explain the concept of Network Address Translation (NAT) and its significance in modern computer networks. [4]

(b) Discuss the various methods used for IPv4 to IPv6 transition in modern networks. Analyze the advantages and disadvantages of each method in terms of implementation complexity, performance impact, and compatibility with existing IPv4 infrastructure. [4]

(c) Discuss HTTPS handshaking process using suitable diagram. [4]

7. a) Why multiplexing is used in data communication? State the merits and demerits of various multiplexing techniques. 3

b) Briefly describe the frequency division multiplexing and de-multiplexing processes. 4

c) State the analogy of communicating with CDMA. 2

d) Why multiple access protocols are needed? Differentiate between CSMA/CD and CSMA/CA 3

3rd year 1st Semester B.Sc. (Hons.) Final Examination, 2023

Duration: 3 hrs. **Full Marks: 60**

Course Code: ICT 3105 **Course Title: ICT Business Analytics and Data Visualization**

Do not write anything on the question paper.

There are 7 **(Seven)** questions. Figures in the right margin indicate marks.

Part – A: Mandatory to answer

1. (a) i. How can one claims the machine learning model is good (efficient) or not good? 1 3

ii. Consider the following data table:

Unit (x)	5	7	12	16	20	24
Price (y)	40	120	180	210	240	?

i. Find out the regression line 7

ii. Find the "?". 8

(b) Assume that there is a csv file, where the last six month's sale information are available. From the sale.csv file, you will use the column of total price per month. 2

i. Write a python code that you can import the data from csv file.

Draw a histogram diagram based on the total price/per month using python

(c) Consider five points { X1, X2,X3, X4, X5} with the following coordinates as a $\boxed{\text{two-dimensional}}$ sample for clustering: X1 = (0,2.5); X2 = (0,0); X3 = (1.5,0); X4 = (5,0); X5 =(5,2)Compose the K-means partitioning algorithm using the above data set. 3

(d) Fill the blanks: 2

Model Building
~~pad~~ ~~Data preparation~~ training
Model Testing
? Validation
Applying Model
? Deployment

(e) How do you handle missing or corrupted data in a Dataset (using application tool python)? 2

Part – B: Answer any 4 (Four) questions.

2. (a) i. Define data analytics and state the steps of the data analysis process. 3

ii. Deduce the applications of data cleaning.

(b) i. "Earth Data (Bu NASA)": mention the type/category of the mentioned data set including your justification. 3

ii. Explain the second party data set using certain examples. → 1st, 3rd

(c) Based on the given data set {1, 3, 6, 7, 12}. 4

i. Find out the standard deviation

ii. Variance

Find the range and coefficient of range of the given data set

(d) How do training and testing data work in Machine Learning? 2

3. (a) i. Using an example, describe the function of outlier. 3

ii. Is it a good idea to follow a hierarchy of descriptive and predictive analytics before applying prescriptive analytics?

(b) i. How can analytics aid in objective decision making? 4

ii. Define support vector machine (SVM) including the types of SVM. → Transaction

(c) i. Short notes on OLAP and OLTP. → Online AnalyticalProcessing 3

ii. Define Big data with certain applications.

(d) How can you distinguish supervised and unsupervised learning in machine learning model? 2

2nd

4\. (a) Provide detail SWOT analysis for a joint-venture of IT (Software Company) business. 4

(b) **Given:** 4

Discount Rate: 10%

Costs:

Y0: 1,00,000 BDT

Y1 - 3: 20,000 BDT

Benefits:

Y0: 0 BDT

Y1 - 3: 1,60,000 BDT

i. Find out ROI.

4

(c) Create a Sphere Model of a project of purchasing smart phone for the "PQS" College students. 3

2nd part (Chapter)

(d) Write the tasks of Data Mining using certain examples. 2

5\. (a) i. Discuss the Data Marts.

ii. Deduce the major characteristics and objectives of the data mining. 3

(b) i. How is training data employed to teach ML and AI models?

ii. Give an example of image annotation. 3

(c) Are over-fitting and under-fitting cases good for machine learning algorithm? According to your answer, justify in your words. 2

d) Let, there is a linear regression model that predicts *Car prices* based on the model of the *car*. 4

10 data points are given with the following actual prices (x) and predicted prices (x_p):

8.

x = [100, 150, 200, 250, 300, 350, 400, 450, 500, 550]

x_p = [110, 140, 180, 240, 290, 320, 380, 420, 480, 520]

i. Find out Root mean squared error

ii. Find out Mean absolute error — 190

iii. Find out R squared → problem

SSR — Sum of square Regression

$\Sigma(x_p - x_{mean})^2$

6\. (a) What is classification? What are the differences between classification and clustering? 3

SST, SSE

(b) Using certain examples, explain lazy learner and eager learner. 1

(c) Evaluate the following table: 4

8.

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

i. Using KNN algorithm, find out (20, 35, ?), where given K=5.

d) Define linear regression and write the properties of linear regression. 2

7\. (a) For example, the web-server installation was ¾ th completed by the end of week 1. The rate of performance would be 75% because by the end of week 1, the planned schedule reflects that the task should be 100 percent complete and only 75 percent of that work has been completed. If planned value is 20000 for the week-1, actual cost is 25000, then find out Earned value, Cost variance, Schedule variance, CPI (cost performance index), and SPI (schedule performance index). 3

(b) How does KNN work? Given an example and explain in your words. 2

(c) i. Define hamming distance using an example. 4

ii. Let it needs to cluster 7 observations into 3 clusters using the algorithm of K-Means clustering. After the first iteration of clusters:

C1, C2, and C3 has the following observations:

C1: {(2,2), (4,4), (6,6)}

C2: {(0,4), (4,0)}

C3: {(5,5), (9,9)}

What is the Manhattan distance for observation (9, 9) from cluster centroid C1 in second iteration?

(4,4) |9-4| + |9-4|

= 5+5 = 10

(d) Explain how the regression process is used in data analysis. List the characteristics of linear regression. 3

Institute of Information Technology

Jahangirnagar University

3rd Year 1st Semester B.Sc. (Hons.) Final Examination- 2023

Duration: 3 hrs. | **Full Marks: 60**

Course Code: ICT-3107 | **Course Title: Information & Data Security**

Do not write anything on the question paper.

There are two parts in the question paper. **Answer from Part A is mandatory. From Part B, answer any 04 (FOUR) questions.** Figures in the right margin indicate marks.

Part: A

Note: Part A contains 12 single-word answer questions. Answer **ALL** of them.

Question 1: Answer **ALL** of the following questions within a single word. 1 x 12 = 12

i. There are many tools for information security. List any one of them.

ii. Which one between **Confidentiality** and **Integrity** is used for protecting information from unauthorized access?

iii. Based on the security goals, how many groups of security attacks there are?

iv. Which one between **Cryptography** and **Digital Signature** is used to authenticate a message?

v. When an attacker impersonates somebody else, it is called __. **(masquerading/ modification).**

vi. Which one between Active attack and Passive attack is normally easier to detect than to prevent?

vii. Which security mechanism is termed as the art of secret writing?

viii Into how many categories, steganography can be divided depending on the nature of the cover media?

ix. Which key is used to encrypt a message using asymmetric-key cryptography?

x. Give an example of a security attack that threatens the integrity of information.

xi. List a common source of invisible ink used in steganography.

xii. Give an example of symmetric-key cryptosystem.

Part: B

Note: Part B contains 6 descriptive-answer questions. Answer any **FOUR (4)** of them.

2. a) Distinguish between passive and active security attacks. Identify which of the following are threats for confidentiality, integrity and availability: (3)
Replaying, Traffic analysis, DoS, Modification, Snooping, Masquerading, DDoS, Repudiation

b) Explain how do the CIA triad and the DAD triad interrelated in assessing and mitigating cyber security risks? (3)

c) Define database security. List some technologies that ensures database security. (3)

d) Draw a general model for information security with necessary directions. List some critical characteristics of information security. (3)

3. a) Give a general idea behind symmetric-key cryptography. How does asymmetric-key cryptography prove the authenticity of the message originator? (3)

b) Briefly describe four security needs provided by cryptography. (3)

c) What are the merits and demerits of asymmetric-key cryptography over symmetric-key cryptography? (3)

d) Three pass protocol can be used to send sensitive information across an insecure network. Give a postal analogy. List some widely used cryptosystem. (3)

4. a) What does authentication mean? A message can be authenticated either by message authentication code (MAC) or cryptographic hash function. Illustrate any one of them. (3)

b) List various categories of authentication factors. What do you mean by two-factor authentication? (3)

c) What are the various approaches used to verify an entity using password? Illustrate any one approach of them. (3)

d) List the possible attacks that may be applied while verifying using user id and password. What are the benefits of one-time password instead of fixed password for verification? (3)

5. a) Define 'risk' and 'risk management' in terms of cyberspace. State the purposes of risk management. (3)

b) Differentiate between positive risk and negative risk. State some strategies used to deal positive and negative risks on your organization. (3)

c) Mention the steps involved in risk management. Briefly describe any one step. (3)

d) How can inference control mechanisms be integrated into cryptographic systems to prevent unauthorized users from deducing sensitive information from encrypted or aggregated data, and what challenges arise in balancing data utility with privacy protection? (3)

6. a) Define digital signature. Illustrate the process of signing and verification used in digital signature. (3)

b) Briefly describe three purposes that are served by a digital signature. (3)

c) Differentiate between conventional signature and digital signature. (3)

d) Let A = {A, B, C, . . . , X, Y, Z, #, $, *} and the corresponding index value of the set A is {i: 1, 2, 3, 4, 5, 6, ..., 26, 27, 28, 29}. Please choose $t = 4$ (for sampling) for each set of partitioned message. (3)

Choose e = (P1, P2, P3, P4). Where, P1: $i = i^2+2$; P2: $i = 2i$; P3: $i = i+1$; P4: $i = i + 3$

Under these circumstances,

P1 maps *first letter* of each partitioned message,

P2 maps *second letter* of each partitioned message,

P3 maps *third letter* of each partitioned message,

P4 maps *forth letter* of each partitioned message.

If message (m) = ABCBEFBAC

Find out, corresponding cipher-text. [Ignore blank space]

7. a) A monetary company is responsible for potential attacks using steganography to hide malicious data through image files. After a most recent breach, the response team found that an attacker had managed to sneak past their defenses by hiding a keylogger inside a legitimate image. (4)

However, the attacker has knowledge of the organization's steganography detection techniques. Under this circumstance, which method of steganalysis would likely be the most effective in detecting such a steganographic attack in the future? Explain in your word.

b) Assume that, the following message needs to be hidden: (6)

"HAB".

The size message is of 3-bytes. Therefore, pixels required to encode the data is 3 x 3 = 9. Let a 4 x 3 image where total pixels are 12 such as:

[(37, 54, 154), (238, 214, 184), (174, 246, 250), (149, 95, 232),
(188, 156, 169), (71, 167, 127), (132, 173, 97), (113, 69, 206),
(255, 29, 213), (53, 153, 220), (246, 225, 229), (144, 72, 165)]

- Find out the encrypted/encode pixel value (after hiding HAB)

c) Using an example, deduce the differences between Cryptography and steganography. (2)

Institute of Information Technology

Jahangirnagar University

3rd year 1st Semester B.Sc. (Hons.) Final Examination, 2023

Duration: 3 hrs. **Full Marks: 60**

Course Code: ICT 3109 **Course Title: Principles of Economics**

Do not write anything on the question paper.

There are 7 (Seven) questions. Figures in the right margin indicate marks.

Part – A: Mandatory to answer

1. (a) What is Economics? Differences between micro and macroeconomics. Why do you study economics as a student of IIT discipline? 3

   (b) Define GDP (Gross Domestic Products) and GNP (Gross National Product). 3

   (c) Define price elasticity of demand. A company used to sell its product at Tk.20 per unit, the market demand was 150 units. When the price was increased to Tk.35 per unit, its demand decreased 110 units. Calculate the price elasticity of demand. 6

Part – B: Answer any 4 (Four) questions.

2. (a) What is the meaning of demand in economics? What are the determinants of demand? 3

   (b) Describe the law of demand and prove the demand law with figure. 5

   (c) Difference between shift in demand and movement along the demand curve with figure. 4

3. (a) What are the components of GDP (Gross Domestic Product)? Explain 4

   (b) Draw and briefly explain the circular flow diagram of GDP. Identify the parts of the model that correspond to the flow of goods and services and the flow of taka for each of the following activities: 8

   i) Ami pays a storekeeper tk. 200 for a quart of milk.

   ii) Rimu earns tk.200 per hour working at a first food restaurant.

   iii) Simu spends tk.1000 to see a drama at the theatre house in Green Road.

   iv) Sharif earns tk.15,000 from his 10 percent ownership of Acme Industrial.

4. (a) What is aggregate demand curve and aggregate supply curve? 3

   (b) Why does the aggregate demand (AD) curve's slope downward? Explain any two reasons 5

   (c) Explain with appropriate figures what happen of price level and output level through aggregate demand (AD) curve: 4

   i. If investment expenditures increase

   ii. If net exports decrease

5. (a) What are the differences between GDP deflator and Consumer price index (CPI)? 2

(b) Below are some data on three goods produced in the economy. 8+2

Year	Price of rice(/kg) in Tk.	Quantity of rice (kg)	Price of cloth (/yard) in Tk.	Quantity of cloth (/yard)	Price of flour(/kg) in Tk.	Quantity of flour (kg)
2015	10	150	12	250	15	250
2016	15	300	20	500	20	300
2017	20	450	25	650	22	350

i) Compute nominal GDP, real GDP, GDP deflator and economic growth rate for each year, using 2015 as the base year.

ii) Explain the economic meaning of the above results.

6. (a) What is opportunity cost? Give one example. 2

(b) What are the factors of production? Explain. 6

(c) Explain production possibilities frontier (PPF) with example. 4

7. (a) Define inflation. Distinguish between demand-pull and cost-push inflation. 6

(b) What are the causes of inflation in Bangladesh? Explain. 6